

Automated Painting Rover (APR)

Competitor Technology Deep Dive — MYRO, ConBotics, DaFang AI, and the RBTX Marketplace

Ref. No.: INC25ETS084848 | **Prepared:** 6 September 2026 | **Method:** Live browsing of each company's own site plus cross-verification via patents, press coverage, and funding databases

Sourcing note: Every technical claim below is followed by a source marker; the full list with URLs is in Section 6. Company marketing claims are labelled as such.

Headline finding: Three of the four links are genuine wall/surface-finishing robot companies with real, verifiable technology — MYRO (interior painting, Singapore/UAE, built on a granted patent with real sensor counts), ConBotics (interior painting, Berlin, TU Berlin spinout with hard dimensional specs), and DaFang AI (multi-function wall finishing, Shenzhen, government-funded and already deployed at scale in China). **The fourth, RBTX, is not a painting-robot competitor at all** — it is a general industrial robot-arm marketplace operated by igus GmbH, with no wall-painting product in its catalogue. Treating it as a competitor in a review document would be a factual error; its correct use is as a price and component reference, covered in Section 5.

1. MYRO Eternal Robotics · Singapore HQ, Dubai branch

COVERAGE RATE	MAX WALL HEIGHT	PRECISION (VENDOR CLAIM)	SETUP TIME
100 m²/hr	3.1 m	~98%	15–20 min

1.1 What it is

MYRO is an interior wall-painting robot developed by Eternal Robotics (the product is now marketed under the separate brand MYRO International, a subsidiary of Mojay Global Holding). It is controlled from an iPad application named "Jessie": the operator draws or imports a floor plan, marks non-paintable zones such as windows, switchboards and doors, and the robot then paints the marked area on its own. Eternal Robotics reports commercial deployments including a luxury Downtown Dubai tower, with further use reported in Brazil and the United States, and lists Sobha Realty, Tata Housing, ALEC Engineering, Jotun, Oberoi Realty and Cyrela among its associations and partners.

Sources: *myro.bot* (Home, Product pages) [1][2]; *Renewable Matter* [3]; *PaintMag* [4]; *Gineersnow* [5].

1.2 Technology stack — grounded in Eternal Robotics' own granted patent

This is the strongest evidence available for any competitor in this document. Eternal Robotics filed patent WO2021255748A1, "Robotic Assisted Wall Painting Apparatus and Method" (priority date 19 June 2020, granted with a corresponding US patent US12303924B2). It names the inventors, is assigned to Eternal Robotics Private Ltd, and describes the machine down to sensor counts and control logic — this is primary-source engineering detail, not a marketing claim.

The patent describes the robot as built from six functional modules: a microprocessor, the human-machine interface ("Jessie" on iPad), a mobile navigation module, a vertical spray actuation module, a safety module, and an error-diagnosis module. Key mechanisms:

Subsystem	Detail (from the patent)
Locomotion	Differential-drive mobile base with 2 stepper motors driving the wheels; 4 distance-measuring sensors keep the robot parallel to, and at a fixed distance from, the wall
Spray arm	A 2-stage vertical telescopic actuator (pulley-belt linear actuators driven by servo motors) raises and lowers the spray gun; a separate stepper motor rotates the gun; an electromagnetic solenoid switches the spray on and off
Motion profile	Each vertical stroke accelerates over the first 10 cm, holds a constant velocity for the middle of the stroke, and decelerates over the final 10 cm, so the paint deposit stays even; the gun is held at a constant 30 cm standoff from the wall
Obstacle sensing	15 ultrasonic sensors (13 of one type, 2 of a second type) plus 2 LiDAR units are positioned to leave no blind spot around the robot's height, width and length
Non-paintable zones	Doors, windows and switchboards are marked on the floor-plan app beforehand; the solenoid automatically cuts the spray when the gun's stroke crosses a marked zone
Reported speed	Patent text states painting speeds up to 85 sq ft per minute ($\approx 474 \text{ m}^2/\text{hr}$ at that instantaneous rate; realised project throughput is reported separately at $\sim 100 \text{ m}^2/\text{hr}$ average)

Source: Google Patents, WO2021255748A1 [6].

1.3 Reported performance and business claims

Claim	Figure
Productivity vs. manual painting	10× (vendor claim, "as measured in previous projects")
Operating cost reduction	70% (vendor claim)
Project completion speed	60% faster (vendor claim)
Material savings	30% less paint wasted (vendor claim)
Single apartment paint time	~ 4 hours by MYRO vs. up to 4 man-days manually (press report)
Maximum wall height served	3.1 m — explicitly indoor-only in current form

Sources: myro.bot efficiency calculator page [2]; PaintMag interview with developers [4]; automationroboticsarduino.com [7].

What this means for APR: MYRO is a mature, patent-backed, commercially deployed product — but it is fundamentally a **floor-driven interior** robot capped at 3.1 m, with no wall-climbing or exterior capability. It does not compete with APR's exterior, vacuum-adhesion use case; it is evidence that floor-plan-driven autonomous painting is a proven, fundable approach, which strengthens rather than undermines APR's premise.

2. ConBotics MalerRoboter · Berlin, Germany

MAX SPRAY HEIGHT

3.20 m

TOTAL WEIGHT

140 kg

PAINT BUCKET

up to 35 L

RUNTIME

4 hrs

2.1 Company background

ConBotics GmbH was founded by three engineering graduates — Cristian Amaya Gomez, Philipp Heyne and David Franke — who began the project in 2020 at the Technical University of Berlin's Centre for Entrepreneurship, having previously worked at the Fraunhofer Institute; the company was formally incorporated in November 2021. Early

funding came from the Berlin Startup Grant and the EXIST Business Start-up Grant, and the product won the Berlin Brandenburg Innovation Prize in 2022. In August 2025 the company closed a Series A round with strategic investors Uzin Utz SE, Teambuild Construction Group (Singapore) and Follmann Chemie, aimed at scaling production and expanding into the DACH region, Singapore and Japan; Atlas Robotech (a Teambuild subsidiary) is the Singapore distribution partner. As of mid-2026 the company has 22 employees.

Sources: TU Berlin entrepreneurship page [8]; Berlin Partner news [9]; baulinks.de [10]; mmm-owl.de [11]; Tracxn company profile [12].

2.2 The MalerRoboter — verified dimensions and specification

Specification	Value
Footprint	60 × 190 × 80 cm
Maximum spray height	3.20 m
Total weight (without the airless unit)	140 kg — marketed as the world's lightest painting robot in its class
Paint compatibility	Works with any airless spray device and any paint type — not locked to one brand
Paint bucket capacity	Up to 35 litres
Battery runtime	4 hours, with a swappable battery for continuous operation
Recharge time	3.5 hours
Charging voltage	110–240 V (works on both US/Asia and European mains)

Source: conbotics.com, MalerRoboter product page (English and German versions cross-checked) [13][14].

2.3 Business model and go-to-market

Unlike MYRO or DaFang, ConBotics does not primarily sell the robot — in Germany it has operated on a **rental model** since April 2024, specifically to avoid a high upfront cost for painting contractors. A sales model targeting Asia was announced for mid-2026. The company also runs "TechnoMaler," a service that pairs large construction projects with qualified painting contractors who bring the MalerRoboter on site, rather than selling the machine direct to an untrained buyer.

Sources: Tagesspiegel [15]; conbotics.com buying-process page [13].

2.4 Modular platform — more than one robot

The MalerRoboter is one attachment on a shared mobile base ConBotics calls "BauRoboter." The same base is designed to be reconfigured, described by the company as similar to interchangeable Lego attachments, for: wall and ceiling sanding (shipping), automated skim-coating/filling of walls and ceilings ("SpachtelRoboter," announced, not yet shipping), and automated floor coating developed jointly with partner Triflex ("FloorBot," announced, not yet shipping). Technology partners named for the coating side include Wagner (whose airless spray machines are integrated into the MalerRoboter) and paint manufacturer Adler.

Source: conbotics.com, "More Construction Robots" page [16].

What this means for APR: ConBotics is the closest of the three to APR in engineering philosophy — a mobile base plus interchangeable task modules — but it is explicitly an **interior, floor-standing** robot with no wall-climbing or adhesion system; its "spray height" of 3.20 m is reached by a mast, not by climbing. It validates the modular-base approach APR could consider for its own subsystem architecture, but is not a mechanism competitor for vacuum-climbing exterior work.

3. DaFang AI Shenzhen DaFang Intelligent Technology Co., Ltd. · Shenzhen, China

FOUNDED

2019

PATENTS FILED

120+

PATENTS GRANTED

51

MIN. CORRIDOR WIDTH

80 cm

3.1 What it is

DaFang AI (legal name Shenzhen DaFang Intelligent Technology Co., Ltd., founded 2019) builds "Four-in-One" wall finishing robots that combine grinding, skim-coating, sanding and painting in a single machine, sold under the DF033 and DF062 model names, plus a DF033-mini line specialised for painting, disinfecting, or sanding alone. The company describes its robots as "L3-level autonomous construction robots" — a self-driving-car-style autonomy grading — combining automatic driving, cloud computing and intelligent sensors to replace manual surface-treatment labour.

Sources: [dafang.ai Products page](#) [17]; [Medium \(Dafang AI company posts\)](#) [18][19].

3.2 Technology stack

Subsystem	Detail
Core autonomy	Described as "L3 autonomous driving technology" combined with a LiDAR perception system and an AI algorithm that builds a 3D map of the apartment and plans an efficient operation path automatically
Actuation	Proprietary "SCA" (Smart and Flexible Actuator) technology drives the tool head across grinding, skim-coating, sanding and spraying functions
Navigation	Visual autonomous driving combined with laser sensing and 5G connectivity for navigation and remote monitoring
Dust handling	Automatic vacuuming is built in, aimed specifically at the high-dust grinding/sanding environment and the associated health risk (pneumoconiosis) to workers
Fleet/data layer	A cloud platform product line sits alongside the physical robots for monitoring and management across sites
Site adaptability	DF033 is stated to work in corridors as narrow as 80 cm and to automatically detect and avoid doors and windows

Sources: [dafang.ai Products page](#) [17]; [Medium, Wuhan China Construction conference post](#) [18].

3.3 Government recognition, funding, and verified deployments

- **Hong Kong:** DF062 and DF033 both hold pre-approval numbers (PA23-051 and PA24-100) under Hong Kong's Construction Industry Innovation and Technology Fund (CITF), which lets Hong Kong construction firms recover up to 70% of the purchase cost; a January 2026 company post cites up to HK\$3 million available per applicant under a related Innovation and Technology Fund scheme.
- **Singapore:** DF062 and DF033 have also been approved for support under the Building and Construction Authority (BCA) fund.
- **Shenzhen municipal government:** DaFang was named among the first batch of official "construction robot application demonstration enterprises" in Shenzhen, under the city's Robot+ industrial policy.
- **Verified project deployment:** the DF033 was used on the Gemdale Huanwan City project, a large residential complex, cited by the company as a successful case study presented at an official China Construction subcontracting conference.

- **Scale claimed:** partnership agreements with 10+ companies across 9 Chinese provinces, construction projects covering millions of square metres, products already sold into Singapore and Hong Kong, with stated plans to expand into Australia and Japan.
- **IP position:** more than 120 patent applications filed, 51 granted, including 2 invention patents and 9 registered software copyrights.
- **Investment:** a funded company (Tracxn lists institutional investors including Yunqi Partners); ranked 26th among 86 tracked competitors in its category, with Okibo listed as a similar company.

Sources: Medium — Hong Kong/Singapore fund support post [19]; Medium — Shenzhen demonstration enterprise post [20]; Medium — Wuhan conference post [18]; dafang.ai News page [21]; Tracxn company profile [22]; LinkedIn company page [23].

What this means for APR: DaFang is the most industrially mature and government-validated of the three real competitors — it has passed formal government subsidy review in two jurisdictions and has documented paid deployments at scale. Its "L3 autonomous driving + LiDAR + cloud" framing is a useful benchmark for how to describe APR's own perception stack credibly to a funding panel. It remains an **interior-corridor** finishing robot; it does not appear to climb exterior vertical walls, which stays APR's specific white space.

4. RBTX operated by igus GmbH · Cologne, Germany

RBTX is not a wall-painting or construction robot company — it is a general-purpose industrial robot marketplace. Including it in a competitor list as if it built a rival product would be a factual mistake worth correcting before this goes in front of a review panel.

4.1 What it actually is

RBTX.com, launched in 2019 and operated by igus GmbH (a German manufacturer of "motion plastics" — self-lubricating polymer bearings, cables and linear components), is an online marketplace that connects buyers with third-party manufacturers of robot arms, grippers, cameras, controllers and mobile platforms. It lists robot arms from brands such as igus's own ReBeL cobot, Universal Robots, ABB, Dobot, Fairino, EPSON and others, spanning cobots, SCARA arms, delta robots, gantry robots, mobile robots and — as of Hannover Messe 2026 — humanoid robots and AGVs. As of the page reviewed for this document, RBTX listed robot arms ranging from roughly ₹1.76 lakh (a small flat linear robot) up to ₹63 lakh (a 1750 mm reach, 8 kg payload Universal Robots arm), spanning payloads from 0.1 kg to 320 kg and reach from 300 mm to over 3 metres.

Sources: rbtx.in/en-IN/components/robots (live catalogue) [24]; igus press releases [25][26]; The Robot Report [27]; Robotics & Automation News [28].

4.2 Why it appeared in this research, and its correct use for APR

Nothing in the RBTX catalogue is built for wall climbing, vacuum adhesion, or spray painting specifically — it is a components and generic-arm marketplace serving pick-and-place, machine tending, assembly and palletising applications. Its relevance to APR is limited to two legitimate uses:

- **Component sourcing:** if APR's design calls for a small articulated arm to position the spray nozzle (as opposed to a fixed nozzle), RBTX is a real place such an arm could be sourced or benchmarked, with transparent pricing.
- **Price benchmarking:** the listed prices give a useful, independently verifiable reference point for what a generic small robot arm costs in the Indian market today, which supports (rather than replaces) APR's own budget justification.

It should not be listed alongside MYRO, ConBotics or DaFang as a competing product in any panel-facing document; doing so would be an easy and avoidable factual error for a reviewer to catch.

5. Side-by-Side Technology Comparison

Dimension	MYRO	ConBotics	DaFang AI	APR (this project)
Target surface	Interior walls, floor-standing	Interior walls/ceilings, floor-standing	Interior walls/corridors, floor-standing	Exterior vertical walls, wall-climbing
Adhesion / access	None — drives on floor	None — drives on floor, mast reaches 3.2 m	None — drives on floor	Active vacuum (negative-pressure) adhesion
Locomotion	Differential-drive wheels	Mobile base (wheeled)	Mobile base, works in 80 cm corridors	Tracks
Perception	15 ultrasonic + 2 LiDAR sensors	Not publicly detailed	LiDAR + AI mapping ("L3 autonomous")	Camera + YOLO26/YOLO11 object detection
Path planning	User-drawn floor plan, auto-paint zones	Not publicly detailed	Autonomous 3D mapping + path planning	Boustrophedon Cellular Decomposition
Control interface	iPad app ("Jessie")	Not publicly detailed	Cloud platform + app	ROS 2 + web dashboard (Digital Twin/Mission Control)
Max height / reach	3.1 m	3.2 m	Not stated (floor-standing)	Unlimited (climbs the full wall)
Business model	Direct sale / project deployment	Rental (Germany), sale planned (Asia)	Sale, government-subsidised in HK/SG	Not yet determined — prototype phase
Maturity evidence	Granted patent, named commercial projects	Series A funded, rental live since Apr 2024	Government-approved, deployed at scale in China	Concept/requirements phase (this project)

6. Source List

All URLs were accessed on 6 September 2026 via direct site visits (MYRO, ConBotics, DaFang AI, RBTX) and supplementary web research for verification.

1. MYRO — Product page — <https://myro.bot/product>
2. MYRO — Home page (efficiency calculator, partner list) — <https://myro.bot/>
3. Renewable Matter, "Myro Bot Is Changing Internal Wall Painting on Construction Sites" — renewablematter.eu
4. PaintMag, "The Paint Robot of the Future" — paintmag.com
5. Gineersnow, "World's First Wall Painting Robot – MYRO" — gineersnow.com
6. automationroboticsarduino.com, "Wall Painting Robots are the Future of Home Remodeling" — MYRO
7. Google Patents, WO2021255748A1, "Robotic assisted wall painting apparatus and method," assignee Eternal Robotics Private Ltd — patents.google.com/patent/WO2021255748A1/en
8. TU Berlin, Centre for Entrepreneurship, "Nachgefragt: Was macht eigentlich ConBotics?" — tu.berlin
9. Berlin Partner, "Berlin is knorke: Painter robot for the construction industry" — berlin-partner.de
10. baulinks.de, "ConBotics startet mit Industriepartnern internationale Expansion" — baulinks.de
11. mmm-owl.de, "Follmann Chemie und Berliner Startup treiben Robotik im Handwerk voran" — mmm-owl.de
12. Tracxn, ConBotics company profile — tracxn.com
13. ConBotics — MalerRoboter product page (English) — conbotics.com/en/paintingrobot
14. ConBotics — MalerRoboter product page (German) — conbotics.com/de/malerroboter
15. Tagesspiegel, "Berliner Malerroboter malert künftig in Asien" — tagesspiegel.de
16. ConBotics — More Construction Robots page — conbotics.com/en/more-constructionrobots

17. DaFang AI — Products page — dafang.ai/public/product.html
18. Medium (@dafangai), Wuhan China Construction conference post
19. Medium (@dafangai), "DaFang AI receives Fund support from Hong Kong and Singapore"
20. Medium (@dafangai), "DaFang AI becomes the first batch of Shenzhen construction robot application demonstration enterprises"
21. DaFang AI — News page — dafang.ai/news.html
22. Tracxn, Dafang company profile — tracxn.com
23. LinkedIn, DaFang AI company page — linkedin.com/company/dafang-ai
24. RBTX India — Robots for Automation catalogue — rbtx.in/en-IN/components/robots
25. igus Press, RBTX tag archive — press.igus.com/tag/rbtx
26. igus Press, "RBTX robotics marketplace now available in 18 countries" — press.igus.eu
27. The Robot Report, "igus launches new RBTX online shopping experience" — therobotreport.com
28. Robotics and Automation News, "igus showcases humanoid robots and AGVs through RBTX marketplace" — roboticsandautomationnews.com